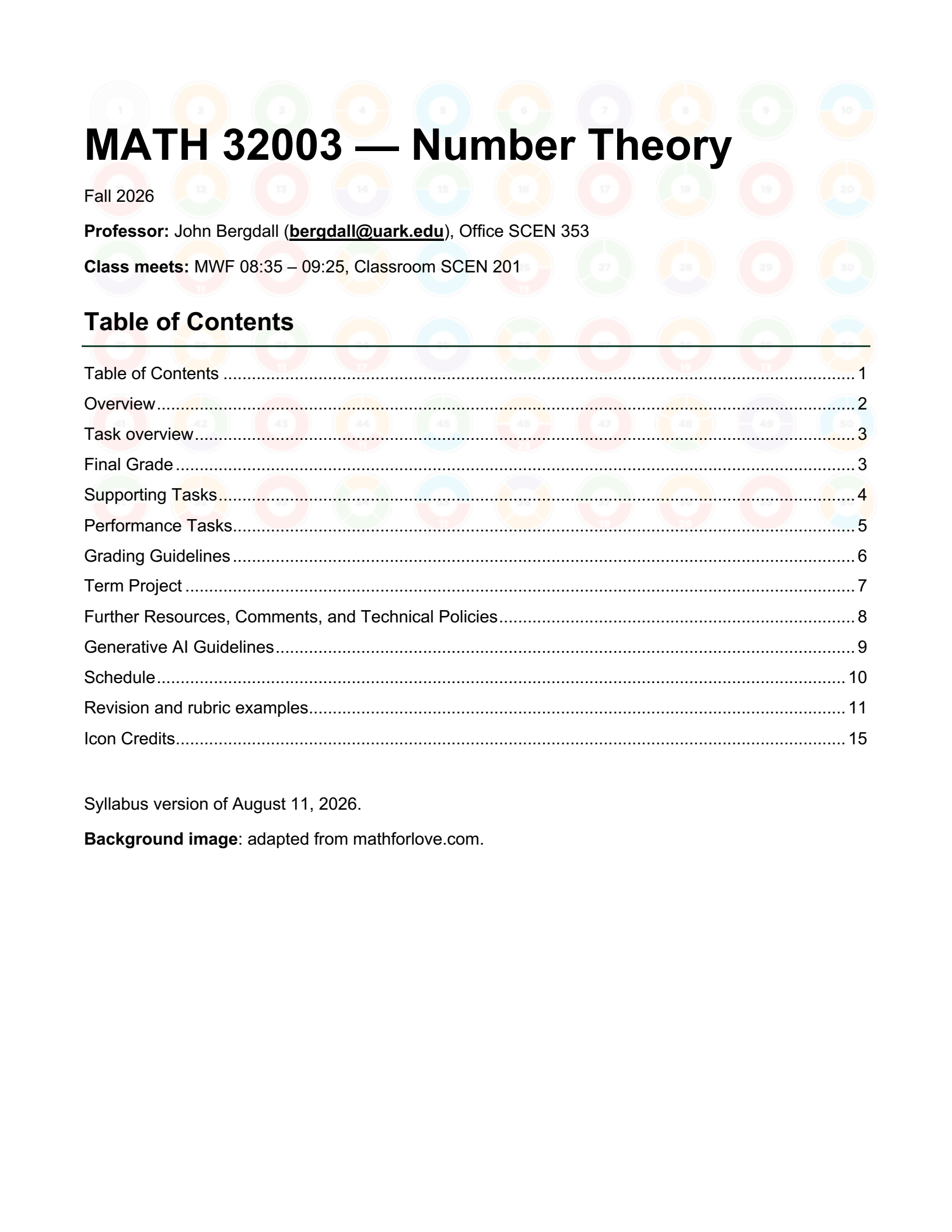




MATH 32003 — Number Theory

Fall 2026

Professor: John Bergdall (bergdall@uark.edu), Office SCEN 353

Class meets: MWF 08:35 – 09:25, Classroom SCEN 201

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Syllabus version of August 11, 2026.

Background image: adapted from mathforlove.com.

Overview

This foundational course on numbers focuses on structure and applications. How are numbers atomically built? How do numbers distribute themselves? Does organizing numbers create useful perspectives for solving problems? How are numbers related to secret codes? We will ask and answer these, and more!

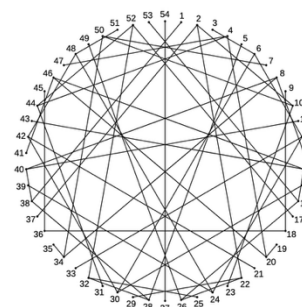
Learning Goals

Learn...	Study...	Explore...	Develop...
foundational number theory problems!	phenomena with modular arithmetic!	algebraic equations and congruences!	applications to secret codes!

Play Time!

Experiment and play are a major focus of this course. To the right is a diagram showing the dynamics of multiplying by **38** in the world of modular arithmetic modulo **54**. What patterns and symmetries do you see?

demonstrations.wolfram.com/ModularMultiplicationOnACircle.



Basic Resources

 Textbook A Friendly Introduction to Number Theory (4th ed.) by Silverman.*	 Course materials Jupyter: <u>Google Colab</u> Homework: <u>Gradescope</u> (Cost: Free)	 Office hours Monday 14:00 – 14:45 Wednesday 13:15 – 14:45 Thursday 14:00 – 14:45
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*If costs cause hardship or would limit your course access, email me for support.

Tips for Success

This class is designed for you to pass. The revision process puts you in control of your grade. Everyone will need some help at some point in the semester. So, here are a few tips and resources you can pursue.

 Message! Write for help.	 Questions! Ask always	 Teamwork! Ask a friend.	 Read! Keep up with class.	 Budget! Plan your time well.
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Task overview

Course Tasks are divided into Supporting Tasks and Performance Tasks. Supporting Tasks keep the class moving. Performance Tasks are where you show understanding. Further details appear below.



Supporting Tasks

Participation, Reading Checks, and Skill Tasks



Performance Tasks

Labs, Mastery Assessments, and Project

Turn-in

Reading Checks will be due shortly before class. Other tasks will be due late Wednesday evenings.

Revisions

Revisions reinforce your learning. Your Reading Checks, Labs, and Mastery Assessments all have revision opportunities.

Late Tasks

Completing Tasks by their due dates keeps you on track. Still, you may turn work in late as part of the revision process (you lose the chance to revise it). You benefit from turning in partial work!

Final Grade

Scores will be weighted according to the percentages below. Those weighted scores will then determine your final grade as follows: A/B/C/D at 90/80/70/60%. I reserve the right to make overall grades *more* lenient.

Task	Weight
Class Participation	5%
Reading Checks	10%
Skill Tasks	10%
Labs	25%
Mastery Assessments	25%
Project	25%

Supporting Tasks

Class Participation (5 points possible)

- **Attendance.** 2 points for better than 80% attendance, 1 point for better than 66% attendance.
- **Introductory survey.** 1 point for responding to the initial course survey.
- **Class welcomes.** 1 point for each of two class welcomes during the semester.

Surveys. We will conduct three surveys throughout the course, in fact. For the second and third surveys, bonus points will be awarded based on total class participation of 80%.

Class welcomes. A class welcome is a 1-2 paragraph summary of the prior class. You work with a partner to write the summary, post it to Blackboard, and read it aloud to start the next meeting.

Reading Checks (1 point per question)

- 6 questions per reading, in a timed and auto-graded check on Gradescope.
- Revisions will include complete reasoning, in writing.

Reading Checks. Multiple choice or select all questions, answered on a computer. You take a check prior to a class meeting, to be prepared for the class discussion. They are open book/notes. You may *not* use generative AI to answer the questions. If you do not get a question initially correct, then revision is possible! Your revision will be writing a detailed response that identifies the correct answer and explains the reasoning.

Reading Check Revisions

Here is how you will “revise” your Reading Checks. See also the example section later.

- **Multiple choice questions.** Identify the correct answer along with an explanation for why it is correct.
- **Select all questions.** First, identify the correct and incorrect options. Second, explain why the correct ones are correct and why the incorrect ones are incorrect.
- **Complete and accurate revisions earn back full points. Partial explanations earn 0-0.5 points.**

Skill Tasks (Graded fully or for completion)

- Short, focused tasks, to help you practice a single technique or computation.
- Turned in electronically. One problem *may* be selected for grading. The rest are for completion.
- Answer keys will be provided, to self-assess learning.

Skill Tasks. These are low-stakes chances to confirm for yourself that you understand a computational or conceptual skill. We will sometimes (with announcement) say we are going to grade one of the problems. In that case we will follow the Task Grading rubric below. Model solutions for Skill Tasks will be provided at the due date. Therefore, no late work will be accepted and no revisions will be possible.

Performance Tasks

Labs

- Focused explorations in Jupyter notebooks, with layered tasks.
- Collaborate! If you attribute credit to your source(s).

The goal of the labs is to give you experience *doing* mathematics. You will have roughly two weeks to complete each lab. A dedicated workday will happen about halfway through the process. The individual tasks in the lab will be graded according to the Task Grading rubric below. You *may* use generative AI to help you code, but you should *not* use it to design and analyze the experiments. (This is literally no fun.)

Lab Revision Specifications

You will be allowed to revise your labs, by re-turning them in, roughly 72 hours after the initial grading is returned. To revise your lab, you will copy and then edit your original notebook, and then I will re-collect it. I will re-assess the work on its own. You do not need to explain your updates or why you made them.

Mastery Assessments

- In-class, 50-minute, written tests: **Sep. 14, Oct. 9, and Nov. 6.**
- Focused on the first three units learning goals.

Your Mastery Assessments will be conducted once per unit. You will know from the start of each unit what topics will be covered. A large portion of the assessment come from your Reading Check and Skill Tasks. The final unit on cryptography is laid back (no written assessment) to allow space for the project.

Mastery Assessment Revisions

You will be allowed to revise your Mastery Assessments by taking a second version of the assessment with conceptually similar problems.

Term Project

- Small groups (2-4 people) working throughout the semester.
- 10-minute presentations on either **Nov. 30** or **Dec. 2.**
- Jupyter notebook due **Dec. 9 at 8:00 AM** (our “final exam” time).

Your Term Project will connect your interests with the course and expand your ability to independently study mathematics. Some steps will be graded on completion. Others will be graded through a more extensive process. Grading guidelines and important due dates are in the Term Project and Schedule sections.

Project Revisions

Project tasks (except the final presentation/notebook) can be revised 72 hours after the initial grading.

Grading Guidelines

Reading Checks

All Reading Check questions are “select all...” or “multiple-choice” questions. You will earn 1 point for the correct answer and 0 for an incorrect one. See above for comments on revising your answers.

Task Grading Rubric

For your Skill Tasks, Labs, and Mastery Assessments, excelling means understanding both concepts and mechanics. It also requires presenting answers in an easy-to-follow fashion, with clear explanations. The following 6-point rubric will be used to assess your solutions.

Criterion	0 points	1 point	2 points
Concept or strategy (2 points)	Answer totally misidentifies or misuses crucial concepts and strategies.	Answer accurately uses at least one crucial concept or tries a productive strategy.	Answer identifies all crucial concepts and succeeds with a productive strategy.
Mechanics or computation (2 points)	Answer has major and/or recurring issues with its mechanics or computation.	Answer shows proficiency in at least some mechanics and computation.	Answer’s mechanics and computations are flawless (of the intended level*).
Grade-ability (1 point)	Grading is impossible (poor writing, wrong problem, ...)	The solution is for the right problem, and the writing is neat enough to assess the work.	
Presentation (1 point)	Solution omits a goal and outcome or major logic.	The solution has a clear goal or outcome, written with complete sentences giving all significant explanations.	

* “...intended level” means you don't simplify the problem to an extreme. See Solution #4 in the worked example below.

Originality

Your written work must be in your own words and reflect your own understanding. If a student turns in an answer that uses approaches, results, terms, etc. that are not consistent with the context of the course, they will receive a “temporary 0” on the Task. The 0 can be removed upon meeting with me to discuss the origin and logic of the solution. See the Generative AI Guidelines section for more detail.

Term Project

Your Term Project will occupy you for most or all of the semester.

Project Steps

1. Think!

Browse the text or search the internet. Create a list of topics.

2. Talk!

Meet with me and peers to discuss hopes and dreams.

3. Plan!

Work in a group to write a clear proposal with concrete goals.

4. Draft!

Create a project outline, including a bibliography.

5. Work!

Finish science, construct a Jupyter notebook, and present.

6. Reflect!

Testify to your teammates' contributions.

Rubric

Step 1 — Brainstorming (1 point)

- Graded for completion.

Step 2 — Discussion (1 point)

- Graded for completion.

Step 3 — Proposal (2-3 paragraphs, 3 points)

- Is a precise and achievable goal proposed?
- Is substantial mathematics proposed?
- Is a significant experiment proposed?

Step 4 — Outline (1-2 pages, 3 points)

- Is there an outline of the notebook's math?
- Is there an outline of the experiment?
- Is there a sufficient bibliography?

Step 5a — Jupyter Notebook (7 points)

- Is the notebook properly formatted?
- Is there an introduction with a clear goal?
- Is the background sound and sufficient?
- Does the body support the goal?
- Is there a significant mathematical quality?
- Is there a significant experimental quality?
- Is there a conclusion with a clear summary?

Step 5b — Presentation (7 points)

- Does everyone sufficiently participate?
- Is the investigation motivated?
- Is the main outcome explained?
- Is the math substance explained?
- Are questions raised and answered?
- Was the presentation overall accurate?
- Was the presentation long enough?

Step 6 — Contribution Testimony (1 page, 3 points)

- How much did you and your teammates contribute to the project?

Further Resources, Comments, and Technical Policies

Appointments and questions

Your questions are important. Office hours may not fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) your availability and (iii) if meeting in person is important. I will try to respond within 24 hours. If I don't, please send a reminder.

Access

I want you to pass this class, with success. If you find resources not provided, tell me and I will work to fix the issue. The Center for Educational Access (cea.uark.edu) coordinates student accommodations. Please request your accommodation letter early in the semester and meet with me to discuss it.

Names and pronouns

You deserve to be called how you want. Tell me your preferred name and pronouns any time. When meeting, re-introduce yourself. Everyone's names, pronouns, etc. should be used respectfully.

Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for me and your peers.

Academic integrity

I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own assignment instructions. When you have questions, ask me rather than assuming you know the answer.

Inclement weather and **Cancellation (emergencies)**

If the University closes due to inclement weather, our course will *not* meet online. Alternative learning materials will be provided. Assessments will be postponed until the next day the University is open.

If class is cancelled on short notice, I will try to find a substitute. If I cannot, alternative learning materials will be supplied. Regardless, an announcement will be sent via email and posted to Blackboard.

Devices

I will record class meetings using in-room recording systems. You will often use your devices (phone, tablet, computer, etc.) to participate in meetings. You are expected to use those devices productively. Non-course activity distracts your peers from learning and me from teaching. It will not be tolerated.

Generative AI Guidelines

Generative AI tools such as ChatGPT are powerful tools for learning. In this course, you are encouraged to responsibly use those tools to better your understanding.

Learning is messy. It has neither start nor finish. AI is just one tool you may use to better understand the material, and it's not going away. Students may use it to summarize or re-explain ideas from their lessons. If you do that in this course, let me know what you find. I designed this course knowing about generative AI capabilities. The revision system is meant to relieve the pressure you may feel to take learning shortcuts.

Basic Expectations

- **Reading Checks.** Complete Reading Checks without directly using generative AI.
- **Skill Tasks and Labs.** Complete these without directly using generative AI.

Here is a more detailed list of my expectations of fair and unfair use of generative AI in this course.

Fair Use

- You may use generative AI before or after reading to reinforce your understanding.
- You may use generative AI to help explain a Reading Check question once it is graded.
- You may use generative AI to generate examples like Skill Tasks or Lab Tasks.
- You may use generative AI to help you find mistakes in graded work.
- You may model your language usage on the usage of a generative AI assistant.

Unfair Use

- You may not directly ask generative AI to answer your Reading Check questions.
- You may not use generative AI to generate revisions to Reading Check questions.
- You may not directly ask generative AI to answer Skill Tasks or Lab Tasks.
- You may not use generative AI to generate revisions to Labs.
- You may not copy outputs from generative AI assistants (or anywhere else).

Temporary 0.

Your written work must be in your own words and reflect your own understanding. If a student turns in an answer that uses approaches, results, terminology, etc. that are not aligned with the context of the course, they will receive a “temporary 0” on the problem. The 0 can be removed upon meeting with me to discuss the origin and logic of the solution. Note: a written revision is not available for a solution I put a temporary 0 on. (Repeated from the Grading Guidelines section, for emphasis.)

Internet searches

Modern internet searches return answers based on generative AI. The above usage extends to internet searches such as Google, Bing, etc.

Schedule

Unit	Dates	Theory Topic	Reading (Silverman)
1	08/19 – 09/14	Introduction to number theory. Mastery assessment: Sep. 14 th . Lab due: Sep. 16 th	Ch. 1-3, 5-8, 30
2	09/16 – 10/09	Congruence arithmetic and primes. Mastery assessment: Oct. 9 th . Lab due: Oct. 7 th .	Ch. 8-12, 41-43
3	10/12 – 11/06	Quadratic reciprocity and beyond. Mastery assessment: Nov. 6 th . Lab due: Nov. 4 th .	Ch. 20-25
4	11/09 – 11/23	Cryptography	Ch. 16-18

Project Due Dates

Step #	Date	Due
1	09/04	Topic brainstorm
2	09/14 – 09/18	Topic meeting and discussion
3	10/02	Project proposal
4	10/30	Project outline
5a	11/30 – 12/02	Presentation
5b	12/09	Jupyter notebook
6	12/09	Contribution testimony

Revision and rubric examples

Reading Check Revisions

Your Reading Checks feature multiple choice and select-all questions. Revising them requires explaining the logic behind the “right” answers and, for select-all questions, the logic behind the “wrong” answers.

RC Revision example 1

Question (Multiple choice). Which of the following indefinite integrals are calculated accurately?

- (a) $\int \sin(x) \, dx = \cos(x) + C$
- (b) $\int \cos(x) \, dx = \sin(x) + C$
- (c) $\int \tan(x) \, dx = \sec^2(x) + C$

Revision (1/1). The integral (b) is accurate by p. 381 of our book. Since this is a multiple-choice question, I don't address (a), (c). *Tip.* You can justify your statement by citing a trusted source, like the textbook.

RC Revision example 2

Question (Select all that apply). Which of the following indefinite integrals are calculated accurately?

- (a) $\int \cos(x) \, dx = \sin(x) + C$
- (b) $\int \sin(x) \, dx = \cos(x) + C$
- (c) $\int \tan(x) \, dx = \sec^2(x) + C$

Revision (1/1). The integral (a) is right but (b), (c) are not. The answer (a) is valid by p. 381 of our book. But (b), (c) are false because p. 381 says $\int \sin(x) \, dx = -\cos(x) + C$ and $\int \tan(x) \, dx = \ln(|\sec(x)|) + C$. *Tip.* To show something is false, you can provide a corrected statement.

RC Revision example 3

Question (Select all that apply). Suppose $y = f(x)$ and $y = g(x)$ are two curves lying over x in $[a, b]$. Which of the following statements are accurate?

- (a) The area between the two curves is always non-negative.
- (b) The integral $\int_a^b [f(x) - g(x)] \, dx$ represents the area between the two curves.
- (c) The integer $\int_a^b [f(x) - g(x)] \, dx$ is always non-negative.
- (d) The integral $\int_a^b |f(x) - g(x)| \, dx$ represents the area between the two curves.

Revision (1/1). The answers (a) and (d) are accurate, but (b) and (c) are not. For (a), areas are always non-negative. For (d), this integral is how you calculate the area (we learned this in the reading). To show by (b) and (c) fail, we produce counterexamples. Suppose $f(x) = x$ and $g(x) = 2$ over the interval $[0, 1]$. Then $\int_0^1 (x - 2) \, dx = 1/2 - 2 = -1.5$. So, (c) fails. In this case, the area is 1.5, not -1.5, which is why (b) is false. *Tip.* To show something is false, provide a counterexample, like we do here for cases (b) and (c).

RC Revision example 4 (Multiple Choice)

Question (Multiple choice). Which of the following indefinite integrals are calculated accurately?

- (a) $\int \sin(x) \, dx = \cos(x) + C$

(b) $\int \cos(x) \, dx = \sin(x) + C$

(c) $\int \tan(x) \, dx = \sec^2(x) + C$

Revision (0/1). The integral (c) is accurate. *Grading logic.* The revision is just wrong.

RC Revision example 5 (Select All)

Question (Select all that apply). Which of the following indefinite integrals are calculated accurately?

(a) $\int \cos(x) \, dx = \sin(x) + C$

(b) $\int \sin(x) \, dx = \cos(x) + C$

(c) $\int \tan(x) \, dx = \sec^2(x) + C$

Revision (0.5/1). The integral (a) is accurate by p. 381 in our text. *Grading logic.* The revision is partially right, but this is a select-all question. You must address (b) and (c) as well.

RC Revision example 6 (Select All, Flawed Reasoning)

Question (Select all that apply). Suppose $y = f(x)$ and $y = g(x)$ are two curves over x lying in $[a, b]$. Which of the following statements are accurate?

- (a) The area between the two curves is always non-negative.
- (b) The integral $\int_a^b [f(x) - g(x)] \, dx$ represents the area between the two curves.
- (c) The integer $\int_a^b [f(x) - g(x)] \, dx$ is always non-negative.
- (d) The integral $\int_a^b |f(x) - g(x)| \, dx$ represents the area between the two curves.

Revision (0.5/1). The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative. For (d), this integral is how you calculate the area (we learned this in the reading). The reason (b) is wrong is because the absolute value in (d) needs to be there. The reason (c) is wrong is that the integral could be negative, if the area between the curves was negative. *Grading logic.* The issue is the final phrase "...if the area between the curves was negative." That just is not an accurate statement, since area must be non-negative. I read revisions closely and points are taken off for inaccurate statements like this. (This is also why I recommend, as above, that you try to provide concrete examples to disprove statements.)

Task Rubric Example

Here we provide an example for how Performance Tasks are graded.

Question. Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

As a grader, I anticipate a student will convert the question into an integral calculation by subtracting one function from the other. Then, they will carry out the integral calculation by finding antiderivatives.

Solution #1

$\cos(x) \geq \sin(x)$ on $[0, \pi/4]$

$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = [\sin(x) + \cos(x)]_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$

Grading Solution #1: 4/6

- **Concept.** 2/2. The solution finds an integral, after subtracting one function from the other.

- **Mechanics.** 1/2. The integral is calculated completely accurately but the inequality is not explained.
- **Grade-ability.** 1/1. The writing (typing) is neat.
- **Presentation.** 0/1. There is no writing that indicates the goal, outcome, or approach of the solution. (This includes the mysterious " $\cos(x) \geq \sin(x)$ " just floating to start the solution.)

Important point: the question asks for an area between two curves. So, the goal is to find an area. The solution just calculates an integral. The missing presentation is the connection between the two!

Solution #2

We find the area by calculating an integral. Since $y = \cos(x)$ lies above $y = \sin(x)$ on $[0, \pi/4]$ (I graphed it), the relevant integral is:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx.$$

$$\text{Now we calculate: } \int_0^{\pi/4} \cos(x) - \sin(x) \, dx = [\sin(x) + \cos(x)]_0^{\pi/4} = 0 + 1 - 1 = 0.$$

Since areas are positive, the area between the curves is 1.

Grading Solution #2: 4/6

- **Concept.** 1/2. The solution calculates an integral, but incorrectly. There is a mechanical point missing there, but I also take off a conceptual point. The reason is that if you truly believe the start of the problem, then you should know *conceptually* there is no way the answer is -1.
- **Mechanics.** 1/2. The integral is not calculated accurately, so the answer is wrong.
- **Grade-ability.** 1/1. The writing (typing) is neat.
- **Presentation.** 1/1. The writing narrates in a clear way how the question is being answered.

This may take getting used to, but conceptual gaps include situations where an answer is inconsistent at the level of pure thought with foundational knowledge surrounding the question.

Solution #3

$$\text{We find the area by calculating the integral } \int_0^{\pi/4} \sin(x) - \cos(x) \, dx = 1 - \sqrt{2} \approx -0.414...$$

Since areas are positive, the answer is 0.414...

Grading Solution #3: 5/6

- **Concept.** 1/2. The solution finds an integral but it has misidentified the difference to integrate and there is no indication the solver realizes the issue.
- **Mechanics.** 2/2. The integral is calculated just fine and within the realm of the calculation that needs to be completed.
- **Grade-ability.** 1/1. The writing (typing) is neat.
- **Presentation.** 1/1. The writing narrates in a clear way how the question is being answered.

You may want to compare with Solution #2. The solutions are similar. Here, the mechanics are perfect and there they are not. In conceptual terms, both solutions have issues. In this #3, you might argue there is more of a conceptual gap. That's okay. My rubric rewards flawless aspects of the work while being generous with mistakes. Similar mistakes do not count double in my grading scheme.

Remember: your professors have agency in how they grade. Grading is not objective, even if someone has told you otherwise. To mitigate that, my rubric is designed to prioritize being transparent and upfront.

Solution #4

The area is given by $\sin(\pi/4) - \cos(\pi/4) = \sqrt{2}/2 - \sqrt{2}/2 = 0$.

Grading Solution #4: 3/6

- **Concept.** 0/2. The solution does not point to a productive strategy to answer the question as given. (Plus, it is a conceptual issue to get area = 0.)
- **Mechanics.** 1/2. The calculation is literally correct but it's overly simple.
- **Grade-ability.** 1/1. The writing (typing) is neat.
- **Presentation.** 1/1. The answer is clear in what it claims has happened.

Revising Tasks

Let me illustrate how I would expect you to revise Solution #1 or #4 above, if you had these on a Mastery Assessment (for instance). If you have any questions when revising your work, please email me to ask!

Revising Solution #1

Original score: 5/6.

When x is in $[0, \pi/4]$ we have $\cos(x) \geq \sin(x)$. To calculate the area, we therefore subtract the smaller from the larger and determine the integral over $[0, \pi/4]$. What we get is

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = [\sin(x) + \cos(x)]_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$$

And therefore the area is 0.414...

Grader note: this revision would earn 6/6. I am writing a more words than you might, but the point is that the various steps are connected with words so that the grader can easily follow. Note also... please include your original score!

Revising Solution #4

Original score: 3/6.

First, I should take the difference in the right order. We have $\cos(\pi/4) - \sin(\pi/4) = \sqrt{2}/2 - \sqrt{2}/2 = 0$. I forgot to also calculate $\cos(0) - \sin(0) = 1$. Then I take the second answer and subtract the first. I get the area is 1.

Grader note: this revision would earn 4/6. The solver has included conceptual information that shows they understand there is something going on with calculating differences of functions. However, conceptual and mechanical gaps remain. In particular, there is no integral still, so the calculations are far too simple. Note: you may not earn back full points on Performance Tasks. Note also... please include your original score!

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